National Institute of Technology, Kurukshetra

B.Tech (5th Sem.)

Data Warehousing and Data Mining (Mathematics & Computing)

Mid Sem. Exam- I

1. As a Principal of a School, you would like to characterize the results of students who have attendance no less than 90%; with respect to the students' age, class, and number of subjects being taught. You would like to know the percentage of pass students having such characteristics. Precisely, you are interested in the result of 10th class students, and from rural background. Results should be displayed in a tabular form. Draw the Data Mining Query for the same, explaining all the syntaxes being used. (3)

2. Given two objects represented by the tuples (22, 1, 42, 10) and (20, 0, 36, 8):
   (a)Compute the Euclidean distance between the two objects.
   (b)Compute the Manhattan distance between the two objects.
   (c)Compute the Minkowski distance between the two objects, using h= 3.
   (d)Compute the supremum distance between the two objects.
   (g)How is a quantile-quantile plot different from a quantile plot? (3)

3. Explain the following terms associated with DBScan Algorithm with the help of Example:
   (i) Core Points
   (ii) Border Points
   iii) Directly Density Reachable.
   (iv) Density Reachable.
   (v) Density Connected. (3)

4. When we talk of the term Data Mining, fetching patterns out of raw data is the most crucial task that one has to perform. But when we extract patterns using KDD process, in that case, can we say that all the patterns so obtained are interesting? If not, then what makes pattern interesting? Do you think KDD process generates 'only' the interesting patterns along with 'all' patterns being interesting? Comment on the significance of the term 'interesting' with respect to pattern evaluation. (3)

5. In order to perform the modeling of the data warehouse, a star schema or q a snowflake schema may be used. Considering any realistic example, compare the two models indicating the similarities as well as differences between the two. Also, suggest which of the two models may be considered more effective in real-life problems and why? (3)